

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{4} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{12} m_2 \tilde{\mu} g_2^4 (9 y_e^{i1i2} + 4 y_e^{pi2} \delta_{pi1}) \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \right. \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_e^{pi2} \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \delta_{pi1} - \frac{1}{6} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \\
& \text{LF}_{2,2,0}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{12} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] - \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{2,2,0}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& \frac{1}{12} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_e^{pr} \overline{a_e^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] - \frac{1}{8} \tilde{\mu} y_e^{pr} \overline{a_e^{pr}} \\
& (g_2^2 (7 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + 3 g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& \frac{1}{6} \tilde{\mu} y_e^{pr} \overline{a_e^{pr}} (g_2^2 (6 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{5,1,-2}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \frac{1}{6} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_d^{pr} \overline{a_d^{pr}} (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] - \\
& \frac{1}{8} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} (3 g_2^2 (7 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + 5 g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \frac{1}{2} \tilde{\mu} y_d^{pr} \overline{a_d^{pr}} \\
& (g_2^2 (6 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \frac{1}{12} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (3 g_2^2 (3 y_e^{i1i2} - y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{2,2,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} y_u^{pr} \overline{a_u^{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] - \\
& \frac{1}{24} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (3 g_2^2 (3 y_e^{i1i2} - y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] - \frac{3}{4} \tilde{\mu} g_2^2 y_e^{i1i2} y_u^{pr} \overline{a_u^{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \frac{1}{12} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} \\
& (3 g_2^2 (6 y_e^{i1i2} + y_e^{si2} \delta_{si1}) - g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{3,1,0}[\mathbf{m}_u^r, \mathbf{m}_q^p] - \\
& \frac{1}{12} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (3 g_2^2 (3 y_e^{i1i2} - y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{3,2,-1}[\mathbf{m}_u^r, \mathbf{m}_q^p] - \frac{1}{4} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} \\
& (3 g_2^2 (6 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[\mathbf{m}_u^r, \mathbf{m}_q^p] + \\
& \frac{1}{2} \tilde{\mu} y_u^{pr} \overline{a_u^{pr}} (g_2^2 (6 y_e^{i1i2} + y_e^{si2} \delta_{si1}) + g_1^2 (y_e^{i1i2} + y_e^{si2} \delta_{si1} - 2 y_e^{i1s} \delta_{si2})) \\
& \text{LF}_{5,1,-2}[\mathbf{m}_u^r, \mathbf{m}_q^p] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 y_e^{i1i2} \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_1] + \frac{7}{8} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] - \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 (g_2^2 (12 y_e^{i1i2} + y_e^{pi2} \delta_{pi1}) + g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_1] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 (g_2^2 (6 y_e^{i1i2} + y_e^{pi2} \delta_{pi1}) + g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_1] + \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^4 (9 y_e^{i1i2} + 2 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_2] + \\
& \frac{1}{24} m_2 \tilde{\mu} g_2^4 (57 y_e^{i1i2} + 8 y_e^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] - \\
& \frac{1}{8} m_2 \tilde{\mu} g_2^2 (g_2^2 (36 y_e^{i1i2} + 7 y_e^{pi2} \delta_{pi1}) + 3 g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \\
& \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_2] - 2 m_2 \tilde{\mu} g_2^4 y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \\
& \frac{1}{2} m_2 \tilde{\mu} g_2^2 (g_2^2 (6 y_e^{i1i2} + y_e^{pi2} \delta_{pi1}) + g_1^2 (y_e^{i1i2} + y_e^{pi2} \delta_{pi1} - 2 y_e^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_2] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \mathbf{m}_l^p] + \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \mathbf{m}_l^p] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \mathbf{m}_l^p] - \\
& \frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 y_e^{i1i2} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \mathbf{m}_l^{i1}] + 2 m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \tilde{\mu}] - \\
& m_1 \tilde{\mu} g_1^4 y_e^{i1i2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_e^{i2}, \tilde{\mu}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \mathbf{m}_l^p, \mathbf{m}_e^{i2}] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pr}} y_e^{$$